

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0195

Evergreen Park 230 kV
August 20, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0195. The Interconnection Customer proposes a 1455.7 MW natural gas combined cycle facility interconnecting at the Evergreen Park 230 kV substation in the Illinois area. The request was received on January 09, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 136 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$106,164,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: WAYNE TRACE to DARLINGTON RD 100kV
Contingency: RICHLAND to DEKALB SUBSTATION 500kV
Loading: 202.7% (Rating: 111 MVA)
DFAX: 6.3%
Pre-project loading: 120.5%
2. Facility: SPRING CITY to EVERGREEN PARK 230kV
Contingency: EVERGREEN PARK to FISK ST 230kV line
Loading: 198.9% (Rating: 433 MVA)
DFAX: 55.1%
Pre-project loading: 13.6%
3. Facility: HILLCRET to SEGS 5 100kV
Contingency: RICHLAND to DEKALB 500kV Line
Loading: 196.2% (Rating: 157 MVA)
DFAX: 7.5%
Pre-project loading: 126.5%
4. Facility: SLOCUM RD to DEKALB 230kV
Contingency: RICHLAND to DEKALB 500kV Line
Loading: 188.6% (Rating: 508 MVA)
DFAX: 26.9%
Pre-project loading: 111.5%
5. Facility: SLOCUM RD to MCNEIL 230kV
Contingency: DEKALB to RICHLAND 500kV
Loading: 173.9% (Rating: 536 MVA)
DFAX: 26.9%
Pre-project loading: 100.9%

6. Facility: REX BROWN to MONTIBELLO 100kV
Contingency: DEKALB to JONES CHAPEL 500kV line
Loading: 153.0% (Rating: 132 MVA)
DFAX: 5.6%
Pre-project loading: 91.3%
7. Facility: LIGONIER to CYPRESS CREEK 100kV
Contingency: DEKALB to JONES CHAPEL 500kV line
Loading: 152.6% (Rating: 132 MVA)
DFAX: 5.6%
Pre-project loading: 91.0%
8. Facility: KENDALLVILLE to REX BROWN 100kV
Contingency: DEKALB to JONES CHAPEL 500kV
Loading: 152.6% (Rating: 123 MVA)
DFAX: 5.6%
Pre-project loading: 86.5%
9. Facility: MCNEIL SUBSTATION to BREED 230kV
Contingency: DEKALB to RICHLAND 500kV
Loading: 150.5% (Rating: 638 MVA)
DFAX: 26.9%
Pre-project loading: 89.1%
10. Facility: SIDNEY A MURRAY JR HYDROELECTRIC SUBSTATION to BREED 230kV L
Contingency: DEKALB to RICHLAND 500kV
Loading: 150.5% (Rating: 638 MVA)
DFAX: 26.9%
Pre-project loading: 89.1%
11. Facility: FARMERSVILLE to KENDALLVILLE 100kV
Contingency: JONES CHAPEL to DEKALB 500kV
Loading: 150.4% (Rating: 118 MVA)
DFAX: 5.6%
Pre-project loading: 81.6%
12. Facility: PUEBLO XTRA 500-230 kV XFMR
Contingency: BOCA RATON to PUEBLO XTRA 500kV
Loading: 146.0% (Rating: 1,141 MVA)
DFAX: 44.1%
Pre-project loading: 89.7%
13. Facility: CONCORD to BLUEBONNET 230kV
Contingency: PUEBLO XTRA to BOCA RATON 500kV
Loading: 144.9% (Rating: 787 MVA)
DFAX: 29.3%
Pre-project loading: 90.7%
14. Facility: RICHLAND 500/230kV Transformer Bank
Contingency: TROY ENERGY LLC to RICHLAND 500kV
Loading: 142.6% (Rating: 853 MVA)
DFAX: 34.1%
Pre-project loading: 84.3%
15. Facility: STA S VAN NUYS to MINNEHAHA 230kV
Contingency: TROY ENERGY LLC SUBSTATION to RICHLAND 500kV
Loading: 140.5% (Rating: 483 MVA)
DFAX: 17.8%
Pre-project loading: 86.9%
16. Facility: EVERGREEN PARK to FISK ST 230kV
Contingency: SPRING CITY to EVERGREEN PARK 230kV

- Loading: 138.3% (Rating: 585 MVA)
DFAX: 51.0%
Pre-project loading: 11.4%
17. Facility: LAURELES to BIG OAKS 100kV
Contingency: DKALB to JONES CHAPEL 500kV
Loading: 135.1% (Rating: 170 MVA)
DFAX: 5.6%
Pre-project loading: 87.3%
18. Facility: HERALD-TWIN CITIES to PICKFORD 230kV
Contingency: TREGO to BEAER 500kV
Loading: 134.7% (Rating: 664 MVA)
DFAX: 10.5%
Pre-project loading: 111.7%
19. Facility: SIDNEY A MURRAY JR HYDROELECTRIC to MINNEHAHA 230kV
Contingency: RICHLAND to DEKALB SUBSTATION 500kV Line
Loading: 134.6% (Rating: 545 MVA)
DFAX: 17.3%
Pre-project loading: 88.5%
20. Facility: BIG OAKS to CYPRESS CREEK 100kV
Contingency: DEKALB SUBSTATION to JONES CHAPEL 500kV
Loading: 134.2% (Rating: 156 MVA)
DFAX: 5.6%
Pre-project loading: 82.1%
21. Facility: ROBISON PARK to DEKALB 230kV
Contingency: JONES CHAPEL to DEKALB 500kV
Loading: 133.5% (Rating: 985 MVA)
DFAX: 35.8%
Pre-project loading: 80.6%
22. Facility: LIGONIER to MONTIBELLO 100kV
Contingency: DEKALB to JONES CHAPEL SUBSTATION 500kV
Loading: 133.5% (Rating: 159 MVA)
DFAX: 5.6%
Pre-project loading: 82.2%
23. Facility: DEKALB 500 kV/1 kV Transformer Bank
Contingency: RICHLAND to DEKALB SUBSTATION 500kV line
Loading: 133.5% (Rating: 969 MVA)
DFAX: 40.9%
Pre-project loading: 72.1%
24. Facility: HERALD-TWIN CITIES to HIGHGATE 230kV
Contingency: MASON ST 500 kV/230 kV Transformer Bank
Loading: 131.3% (Rating: 518 MVA)
DFAX: 7.6%
Pre-project loading: 110.0%
25. Facility: ROBISON PARK to RUSSELL FORK 230kV
Contingency: DEKALB to JONES CHAPEL SUBSTATION 500kV L
Loading: 127.3% (Rating: 1,157 MVA)
DFAX: 45.7%
Pre-project loading: 69.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WAYNE TRACE to DARLINGTON RD 100kV	Rebuild line with higher-capacity conductor	\$2,013,000	17% (\$343,000)
SPRING ITY to EVERGREEN PARK 230kV line	Rebuild line with higher-capacity conductor	\$9,158,000	100% (\$9,158,000)
HILLCRET to SEGS 5 100kV	Rebuild line with higher-capacity conductor	\$3,520,000	17% (\$600,000)
SLOCUM RD SUBSTATION to DEALB 230kV	Rebuild line with higher-capacity conductor	\$44,746,000	17% (\$7,719,000)
MCNEIL to SLOCUM RD 230kV	Rebuild line with higher-capacity conductor	\$12,666,000	17% (\$2,185,000)
REX BROWN to MONTIBELLO 100kV	Rebuild line with higher-capacity conductor	\$8,188,000	16% (\$1,313,000)
LIGONIER to CYPRESS CREEK 100kV	Rebuild line with higher-capacity conductor	\$10,594,000	16% (\$1,699,000)
KENDALLVILLE to REX BROWN 100kV	Rebuild line with higher-capacity conductor	\$1,998,000	16% (\$320,000)
BREED to MCNEIL 230kV line	Rebuild line with higher-capacity conductor	\$18,176,000	17% (\$3,135,000)
SIDNEY A MURRAY JR HYDROELECTRIC to BREED 230kV	Rebuild line with higher-capacity conductor	\$29,436,000	17% (\$5,078,000)
FARMERSVILLE to KENDALLVILLE 100kV	Rebuild line with higher-capacity conductor	\$7,259,000	16% (\$1,164,000)
PUEBLO XTRA 500-230 kV XFMR	Replace transformer with higher-rated unit	\$28,473,000	19% (\$5,526,000)
CONCORD to BLUEBONNET 230kV L	Rebuild line with higher-capacity conductor	\$22,200,000	15% (\$3,290,000)
RICHLAND 500/230kV Autotransformer	Replace transformer with higher-rated unit	\$34,865,000	17% (\$6,067,000)
STA S VAN NUYS to MINNEHAHA 230kV	Rebuild line with higher-capacity conductor	\$17,356,000	17% (\$2,999,000)
FISK ST to EVERGREEN PRK 230kV	Rebuild line with higher-capacity conductor	\$13,562,000	100% (\$13,562,000)
LAURELES to BIG OAKS 100kV	Rebuild line with higher-capacity conductor	\$4,170,000	16% (\$669,000)
PICFORD to HERALD-TWIN CITIES 230kV	Rebuild line with higher-capacity conductor	\$49,760,000	17% (\$8,624,000)
MINNEHAHA to SIDNEY A MURRAY JR HYDROELECTRIC 230kV Line	Rebuild line with higher-capacity conductor	\$45,494,000	17% (\$7,847,000)
BIG OAKS to CYPRESS CREEK 100kV	Rebuild line with higher-capacity conductor	\$4,515,000	16% (\$724,000)
ROBISON PARK to DEKALB 230kV	Rebuild line with higher-capacity conductor	\$35,741,000	17% (\$6,096,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LIGONIER to MONTIBELLO 100kV	Rebuild line with higher-capacity conductor	\$5,462,000	16% (\$876,000)
DEKALB 500 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$29,407,000	18% (\$5,249,000)
HIGHGATE to HERALD-TWIN CITIES 230kV Line	Rebuild line with higher-capacity conductor	\$23,520,000	17% (\$4,078,000)
ROBISON PARK to RUSSELL FORK 230kV	Rebuild line with higher-capacity conductor	\$46,272,000	17% (\$7,843,000)
	Total allocated to project		\$106,164,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.